Q – 1 Create a Blank window with Tkinter library.

- **Input :-**

```
from tkinter import *

# Main window
root = Tk()

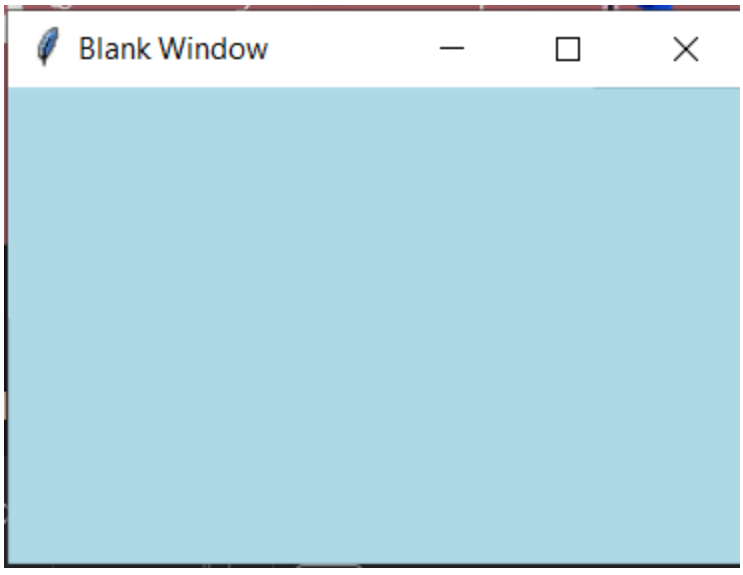
# Window title
root.title("Blank Window")

# Window size
root.geometry("400x400")

# Window background color
root.configure(bg="lightblue")

# Run the application
root.mainloop()
```

- **Output :-**



Q – 2 Create a window with the help of pack() method (add button widgets inside window).

- **Input :-**

```
from tkinter import *

# Main window
root = Tk()

# Window title
root.title("Button Widgets")

# Window size
root.geometry("400x400")

# Window background color
root.configure(bg="lightblue")

# Label
lbl = Label(
   root,
   text="Welcome to Tkinter",
   font=("Arial", 16, "bold"),
   bg="lightblue"
)
lbl.pack(pady=10)

# First button
btn1 = Button(
   root,
   text="Button 1",
   font=("Arial", 12),
   bg="green",
   fg="white"
)
btn1.pack(pady=10)
```

```
# Second button
btn2 = Button(
    root,
    text="Button 2",
    font=("Arial", 12),
    bg="blue",
    fg="white"
)
btn2.pack(pady=10)

# Third button
btn3 = Button(
    root,
    text="Button 3",
    font=("Arial", 12),
    bg="red",
    fg="white"
)
btn3.pack(pady=10)

# Run the application
root.mainloop()
```

- **Output :-**



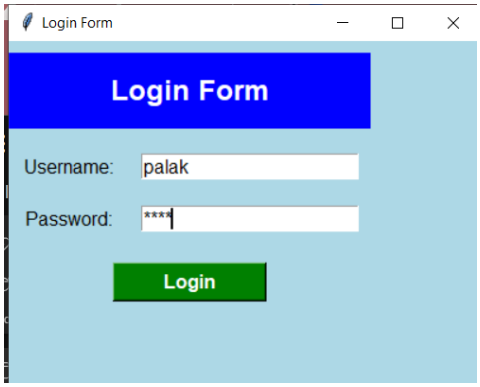
Q – 3 Write a code to create login form using grid() method.

- **Input :-**

```
from tkinter import *
from tkinter import messagebox
root = Tk()
root.title("Login Form")
root.geometry("400x300")
root.configure(bg="lightblue")
lbl = Label(
   root,
   text="Login Form",
   font=("Arial", 18, "bold"),
   fg="white",
   bg="blue",
   width=20,
   height=2
)
lbl.grid(row=0, column=0, columnspan=2, pady=10
username_lbl = Label(
   root,
   text="Username:",
   font=("Arial", 12),
   bg="lightblue"
)
username_lbl.grid(row=1, column=0, padx=10, pady=10)
username_entry = Entry(
   root,
   width=20,
   font=("Arial", 12)
)
username_entry.grid(row=1, column=1, padx=10, pady=10)
password_lbl = Label(
   root,
   text="Password:",
   font=("Arial", 12),
   bg="lightblue"
```

```
)
password_lbl.grid(row=2, column=0, padx=10, pady=10)
password_entry = Entry(
   root,
   width=20,
   font=("Arial", 12),
   show="*"
)
password_entry.grid(row=2, column=1, padx=10, pady=10)
def login():
   username = username_entry.get()
   password = password_entry.get()
   if username == "admin" and password == "1234":
      messagebox.showinfo("Success", "Login Successful")
   else:
      messagebox.showerror("Error", "Invalid Username or Password")
btn = Button(
   root,
   text="Login",
   command=login,
   font=("Arial", 12, "bold"),
   bg="green",
   fg="white",
   width=12
btn.grid(row=3, column=0, columnspan=2, pady=15)
root.mainloop()
```

- **Output :-**



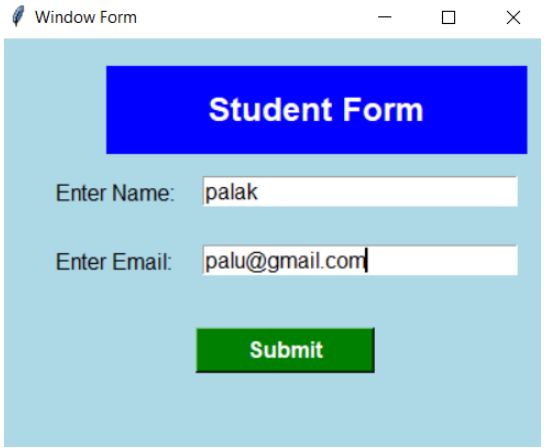
Q – 4 Create a window form with the help of place() method.

- **Input :-**

```
from tkinter import *
# Main window
root = Tk()
root.title("Window Form")
root.geometry("400x300")
root.configure(bg="lightblue")
# Heading label
lbl = Label(
   root,
   text="Student Form",
   font=("Arial", 18, "bold"),
   fg="white",
   bg="blue",
   width=20,
   height=2
)
lbl.place(x=80, y=20)
# Name label
name_lbl = Label(
   root,
   text="Enter Name:",
   font=("Arial", 12),
   bg="lightblue"
)
name_lbl.place(x=40, y=100)
# Name entry
name_entry = Entry(
   root,
   width=25,
   font=("Arial", 12)
)
name_entry.place(x=150, y=100)
# Email label
email_lbl = Label(
```

```
    root,
    text="Enter Email:",
    font=("Arial", 12),
    bg="lightblue"
)
email_lbl.place(x=40, y=150)
# Email entry
email_entry = Entry(
    root,
    width=25,
    font=("Arial", 12)
)
email_entry.place(x=150, y=150)
# Submit button
btn = Button(
    root,
    text="Submit",
    font=("Arial", 12, "bold"),
    bg="green",
    fg="white",
    width=12
)
btn.place(x=145, y=210)
# Run the application
root.mainloop()
```

- **Output :-**



Q – 5 Divide a window form into 2 parts using frame.

- **Input :-**

```
from tkinter import *

# Main Window
root = Tk()
root.title("Window Divided into 2 Parts")
root.geometry("500x350")
root.configure(bg="lightblue")

# Heading
heading = Label(
   root,
   text="Student Registration Form",
   font=("Arial", 18, "bold"),
   bg="lightblue",
   fg="darkblue"
)
heading.pack(pady=10)

# First Frame
frame1 = Frame(
   root,
   bg="white",
   bd=2,
   relief="groove",
   width=220,
   height=230
)
frame1.place(x=20, y=60)

# First Frame Heading
Label(
   frame1,
   text="Student Details",
   font=("Arial", 14, "bold"),
```

```
    bg="white",
    fg="blue"
).place(x=35, y=10)

# Name
Label(frame1, text="Name:", bg="white").place(x=10, y=60)
Entry(frame1, width=18).place(x=75, y=60)

# Roll Number
Label(frame1, text="Roll No:", bg="white").place(x=10, y=100)
Entry(frame1, width=18).place(x=75, y=100)

# Second Frame
frame2 = Frame(
    root,
    bg="lightyellow",
    bd=2,
    relief="groove",
    width=220,
    height=230
)
frame2.place(x=260, y=60)

# Second Frame Heading
Label(
    frame2,
    text="Contact Details",
    font=("Arial", 14, "bold"),
    bg="lightyellow",
    fg="darkgreen"
).place(x=30, y=10)

# Mobile Number
Label(frame2, text="Mobile:", bg="lightyellow").place(x=10, y=60)
Entry(frame2, width=18).place(x=75, y=60)

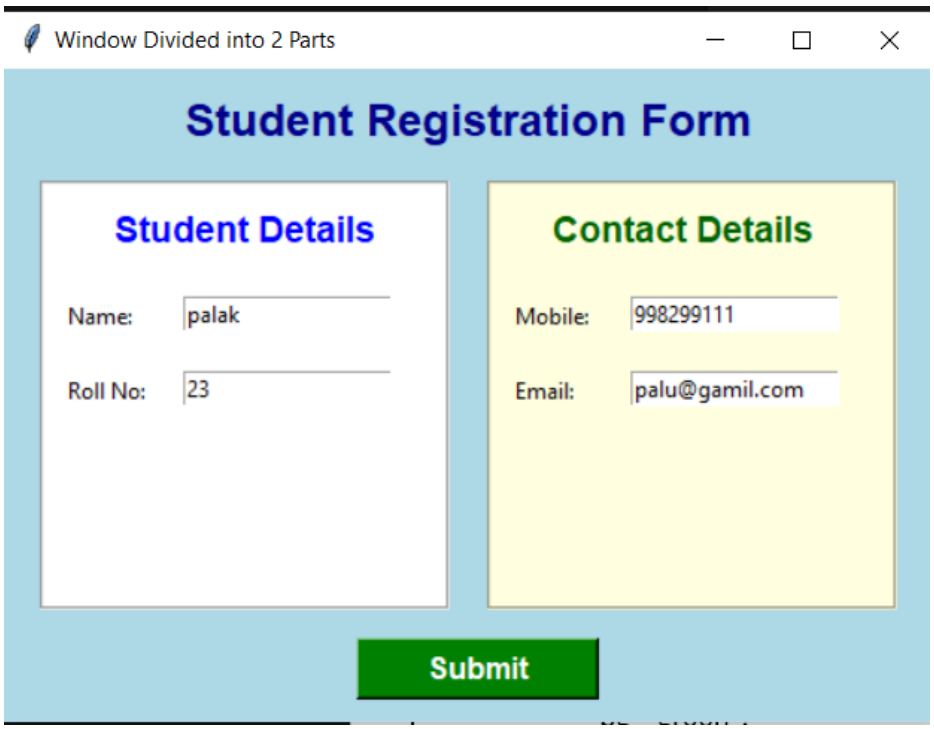
# Email
Label(frame2, text="Email:", bg="lightyellow").place(x=10, y=100)
```

```
Entry(frame2, width=18).place(x=75, y=100)

# Submit Button
Button(
   root,
   text="Submit",
   font=("Arial", 12, "bold"),
   bg="green",
   fg="white",
   width=12
).place(x=190, y=305)

# Run Window
root.mainloop()
```

➢ **Output :-**



Q – 6 Write python code to gather student information using windows gui form using Tkinter widgets. Form Properties: (Name, Email, Phone, Address, Classname(radiobutton), semester(listbox), hobbies(checkbox), image(canvas) & submit button.).

- **Input :-**

```
from tkinter import *
from tkinter import messagebox

# Main Window
root = Tk()
root.title("Student Information Form")
root.geometry("650x700")
root.configure(bg="lightblue")

# Heading
Label(
   root,
   text="Student Information Form",
   font=("Arial", 20, "bold"),
   bg="lightblue",
   fg="darkblue"
).pack(pady=10)

# Name
Label(root, text="Name:", bg="lightblue",
    font=("Arial", 11)).place(x=30, y=70)

name_entry = Entry(root, width=30)
name_entry.place(x=150, y=70)

# Email
Label(root, text="Email:", bg="lightblue",
    font=("Arial", 11)).place(x=30, y=110)

email_entry = Entry(root, width=30)
```

```
email_entry.place(x=150, y=110)

# Phone
Label(root, text="Phone:", bg="lightblue",
    font=("Arial", 11)).place(x=30, y=150)

phone_entry = Entry(root, width=30)
phone_entry.place(x=150, y=150)

# Address
Label(root, text="Address:", bg="lightblue",
    font=("Arial", 11)).place(x=30, y=190)

address_text = Text(root, width=23, height=3)
address_text.place(x=150, y=190)

# Class Name - Radiobutton
Label(root, text="Class Name:", bg="lightblue",
    font=("Arial", 11)).place(x=30, y=260)

class_var = StringVar(value="BCA")

Radiobutton(root, text="BCA", variable=class_var,
        value="BCA", bg="lightblue").place(x=150, y=260)

Radiobutton(root, text="BBA", variable=class_var,
        value="BBA", bg="lightblue").place(x=220, y=260)

Radiobutton(root, text="B.Com", variable=class_var,
        value="B.Com", bg="lightblue").place(x=290, y=260)

# Semester - Listbox
Label(root, text="Semester:", bg="lightblue",
    font=("Arial", 11)).place(x=30, y=300)

semester_list = Listbox(root, height=4, width=15,
                exportselection=False)
```

```
for sem in ["Semester 1", "Semester 2",
        "Semester 3", "Semester 4",
        "Semester 5", "Semester 6"]:
   semester_list.insert(END, sem)

semester_list.place(x=150, y=300)

# Hobbies - Checkbuttons
Label(root, text="Hobbies:", bg="lightblue",
    font=("Arial", 11)).place(x=350, y=300)

reading_var = IntVar()
sports_var = IntVar()
music_var = IntVar()

Checkbutton(root, text="Reading", variable=reading_var,
        bg="lightblue").place(x=430, y=300)

Checkbutton(root, text="Sports", variable=sports_var,
        bg="lightblue").place(x=430, y=330)

Checkbutton(root, text="Music", variable=music_var,
        bg="lightblue").place(x=430, y=360)

# Image - Canvas
Label(root, text="Image Canvas:", bg="lightblue",
    font=("Arial", 11)).place(x=30, y=410)

canvas = Canvas(root, width=150, height=100,
           bg="white", bd=2, relief="sunken")
canvas.place(x=150, y=410)

canvas.create_text(75, 50, text="Student Image",
             font=("Arial", 12), fill="gray")

# Submit Function
def submit():
   name = name_entry.get()
```

```
    email = email_entry.get()
    phone = phone_entry.get()
    address = address_text.get("1.0", END).strip()
    class_name = class_var.get()

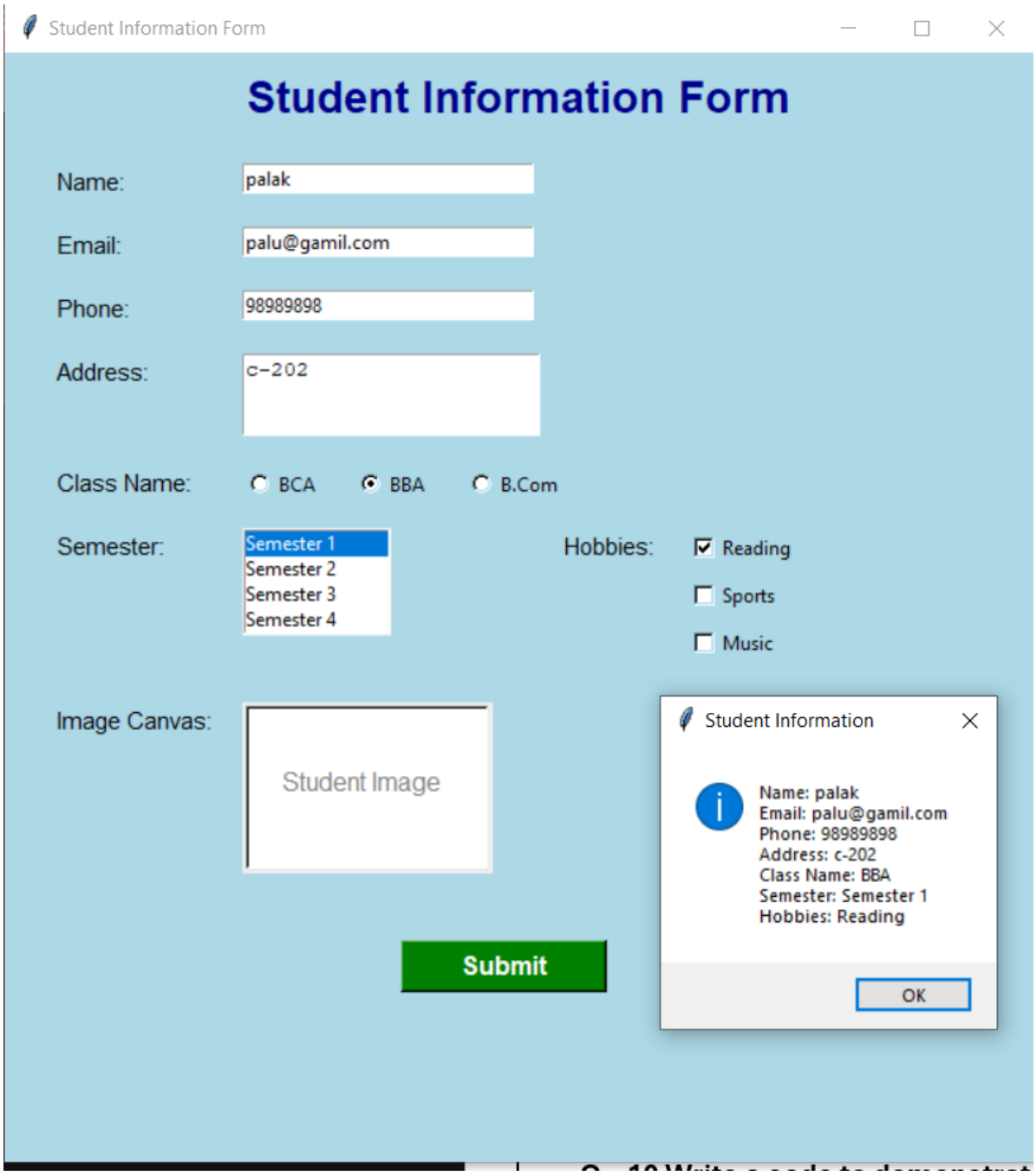
    selected = semester_list.curselection()
    semester = semester_list.get(selected[0]) if selected else "Not selected"

    hobbies = []
    if reading_var.get():
        hobbies.append("Reading")
    if sports_var.get():
        hobbies.append("Sports")
    if music_var.get():
        hobbies.append("Music")

    if not name or not email or not phone:
        messagebox.showwarning(
            "Warning", "Please enter Name, Email and Phone."
        )
        return
    info = (
        f"Name: {name}\n"
        f"Email: {email}\n"
        f"Phone: {phone}\n"
        f"Address: {address}\n"
        f"Class Name: {class_name}\n"
        f"Semester: {semester}\n"
        f"Hobbies: {', '.join(hobbies) if hobbies else 'None'}"
    )
    messagebox.showinfo("Student Information", info)
# Submit Button
Button(root, text="Submit", font=("Arial", 12, "bold"),
       bg="green", fg="white", width=12,
       command=submit).place(x=250, y=560)

# Run Application
root.mainloop()
```

➢ **Output :-**


Student Information Form
Student Information Form
Name:
palak
Email:
palu@gamil.com
Phone:
98989898
Address:
c-202
Class Name:
BCA
BBA
B.Com
Semester:
Semester 1
Semester 2
Semester 3
Semester 4
Hobbies:
Reading
Sports
Music
Image Canvas:
Student Image
Submit
Student Information
Name: palak
Email: palu@gamil.com
Phone: 98989898
Address: c-202
Class Name: BBA
Semester: Semester 1
Hobbies: Reading
OK

Q – 7 Write a code to demonstrate Button Widget in Tkinter:

- **Input :-**

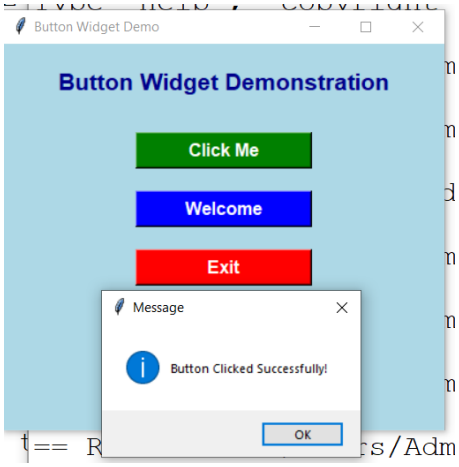
```
from tkinter import *
from tkinter import messagebox

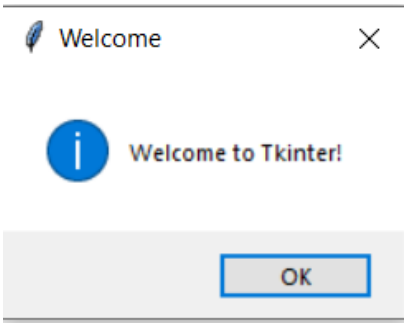
# Main Window
root = Tk()
root.title("Button Widget Demo")
root.geometry("400x350")
root.configure(bg="lightblue")

# Heading Label
heading = Label(
   root,
   text="Button Widget Demonstration",
   font=("Arial", 16, "bold"),
   bg="lightblue",
   fg="darkblue"
)
heading.pack(pady=20)
def click_button():
   messagebox.showinfo("Message", "Button Clicked Successfully!")
def welcome_button():
   messagebox.showinfo("Welcome", "Welcome to Tkinter!")
btn1 = Button(
   root,
   text="Click Me",
   font=("Arial", 12, "bold"),
   bg="green",
   fg="white",
   width=15,
   command=click_button
)
btn1.pack(pady=10)
# Second Button
btn2 = Button(
```

```
    root,
    text="Welcome",
    font=("Arial", 12, "bold"),
    bg="blue",
    fg="white",
    width=15,
    command=welcome_button
)
btn2.pack(pady=10)
btn3 = Button(
    root,
    text="Exit",
    font=("Arial", 12, "bold"),
    bg="red",
    fg="white",
    width=15,
    command=root.destroy
)
btn3.pack(pady=10)
root.mainloop()
```

➢ **Output :-**





Q – 8 Write a code to demonstrate Checkbutton Widget

- **Input :-**

```
from tkinter import *
from tkinter import messagebox

# Main Window
root = Tk()
root.title("Checkbutton Widget Demo")
root.geometry("400x400")
root.configure(bg="lightblue")

# Heading Label
heading = Label(
   root,
   text="Checkbutton Demonstration",
   font=("Arial", 16, "bold"),
   bg="lightblue",
   fg="darkblue"
)
heading.pack(pady=20)

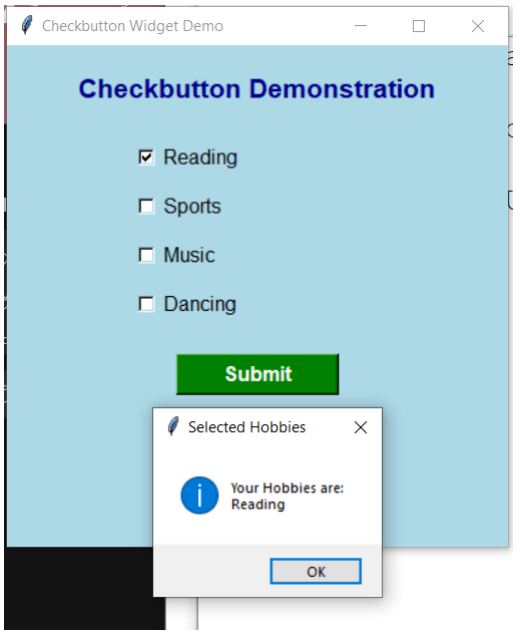
# Variables for Checkbuttons
reading_var = IntVar()
sports_var = IntVar()
music_var = IntVar()
dancing_var = IntVar()
# Reading Checkbutton
cb1 = Checkbutton(
   root,
   text="Reading",
   variable=reading_var,
   font=("Arial", 12),
   bg="lightblue"
)
cb1.pack(anchor="w", padx=100, pady=5)
# Sports Checkbutton
```

```
cb2 = Checkbutton(
    root,
    text="Sports",
    variable=sports_var,
    font=("Arial", 12),
    bg="lightblue"
)
cb2.pack(anchor="w", padx=100, pady=5)
# Music Checkbutton
cb3 = Checkbutton(
    root,
    text="Music",
    variable=music_var,
    font=("Arial", 12),
    bg="lightblue"
)
cb3.pack(anchor="w", padx=100, pady=5)
# Dancing Checkbutton
cb4 = Checkbutton(
    root,
    text="Dancing",
    variable=dancing_var,
    font=("Arial", 12),
    bg="lightblue"
)
cb4.pack(anchor="w", padx=100, pady=5)
def submit():
    hobbies = []
    if reading_var.get() == 1:
        hobbies.append("Reading")

    if sports_var.get() == 1:
        hobbies.append("Sports")
    if music_var.get() == 1:
        hobbies.append("Music")
    if dancing_var.get() == 1:
        hobbies.append("Dancing")
    if hobbies:
```

```
        messagebox.showinfo(
            "Selected Hobbies",
            "Your Hobbies are:\n" + "\n".join(hobbies)
        )
    else:
        messagebox.showwarning(
            "Warning",
            "Please select at least one hobby."
        )
btn = Button(
    root,
    text="Submit",
    font=("Arial", 12, "bold"),
    bg="green",
    fg="white",
    width=12,
    command=submit
)
btn.pack(pady=20)
root.mainloop()
```

- **Output :-**



Q – 9 Write a code to demonstrate Entry Widget

- **Input :-**

```
from tkinter import *
from tkinter import messagebox

# Main Window
root = Tk()
root.title("Entry Widget Demo")
root.geometry("450x350")
root.configure(bg="lightblue")

# Heading Label
heading = Label(
   root,
   text="Entry Widget Demonstration",
   font=("Arial", 16, "bold"),
   bg="lightblue",
   fg="darkblue"
)
heading.pack(pady=20)

# Name Label
name_label = Label(
   root,
   text="Enter Name:",
   font=("Arial", 12),
   bg="lightblue"
)
name_label.place(x=50, y=90)

# Name Entry
name_entry = Entry(root, width=25, font=("Arial", 12))
name_entry.place(x=170, y=90)

# Email Label
email_label = Label(
```

```
    root,
    text="Enter Email:",
    font=("Arial", 12),
    bg="lightblue"
)
email_label.place(x=50, y=140)

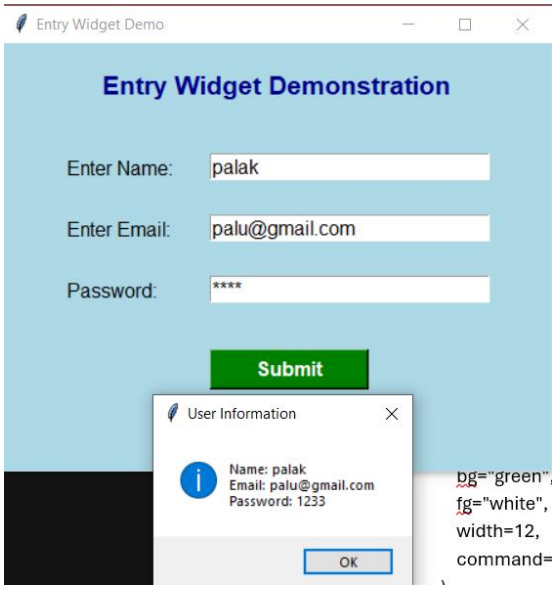
# Email Entry
email_entry = Entry(root, width=25, font=("Arial", 12))
email_entry.place(x=170, y=140)

# Password Label
password_label = Label(
    root,
    text="Password:",
    font=("Arial", 12),
    bg="lightblue"
)
password_label.place(x=50, y=190)

# Password Entry
password_entry = Entry(
    root,
    width=25,
    font=("Arial", 12),
    show="*"
)
password_entry.place(x=170, y=190)
# Submit Function
def submit():
    name = name_entry.get()
    email = email_entry.get()
    password = password_entry.get()
    if not name or not email or not password:
        messagebox.showwarning(
            "Warning",
            "Please fill in all fields."
        )
```

```
        return
    messagebox.showinfo(
        "User Information",
        f"Name: {name}\n"
        f"Email: {email}\n"
        f"Password: {password}"
    )
btn = Button(
    root,
    text="Submit",
    font=("Arial", 12, "bold"),
    bg="green",
    fg="white",
    width=12,
    command=submit
)
btn.place(x=170, y=250)
# Run Application
root.mainloop()
```

- **Output :-**



Q – 10 Write a code to demonstrate Frame Widget

- **Input :-**

```
from tkinter import *
from tkinter import messagebox

# Main Window
root = Tk()
root.title("Frame Widget Demo")
root.geometry("500x400")
root.configure(bg="lightblue")

# Heading Label
heading = Label(
   root,
   text="Frame Widget Demonstration",
   font=("Arial", 18, "bold"),
   bg="lightblue",
   fg="darkblue"
)
heading.pack(pady=15)

# First Frame - Student Details
frame1 = Frame(
   root,
   bg="white",
   bd=2,
   relief="groove",
   width=440,
   height=120
)
frame1.pack(padx=20, pady=10)
frame1.pack_propagate(False)

# First Frame Heading
Label(
   frame1,
```

```
    text="Student Details",
    font=("Arial", 14, "bold"),
    bg="white",
    fg="blue"
).grid(row=0, column=0, columnspan=2, pady=5)

# Name Label and Entry
Label(frame1, text="Name:", bg="white").grid(
    row=1, column=0, padx=10, pady=5
)

name_entry = Entry(frame1, width=25)
name_entry.grid(row=1, column=1, padx=10, pady=5)

# Roll Number Label and Entry
Label(frame1, text="Roll No:", bg="white").grid(
    row=2, column=0, padx=10, pady=5
)

roll_entry = Entry(frame1, width=25)
roll_entry.grid(row=2, column=1, padx=10, pady=5)

# Second Frame - Contact Details
frame2 = Frame(
    root,
    bg="lightyellow",
    bd=2,
    relief="groove",
    width=440,
    height=100
)
frame2.pack(padx=20, pady=10)
frame2.pack_propagate(False)

# Second Frame Heading
Label(
    frame2,
    text="Contact Details",
```

```
    font=("Arial", 14, "bold"),
    bg="lightyellow",
    fg="darkgreen"
).grid(row=0, column=0, columnspan=2, pady=5)

# Email Label and Entry
Label(frame2, text="Email:", bg="lightyellow").grid(
    row=1, column=0, padx=10, pady=5
)

email_entry = Entry(frame2, width=25)
email_entry.grid(row=1, column=1, padx=10, pady=5)

# Submit Function
def submit():
    name = name_entry.get()
    roll = roll_entry.get()
    email = email_entry.get()

    if not name or not roll or not email:
        messagebox.showwarning(
            "Warning",
            "Please fill in all fields."
        )
        return

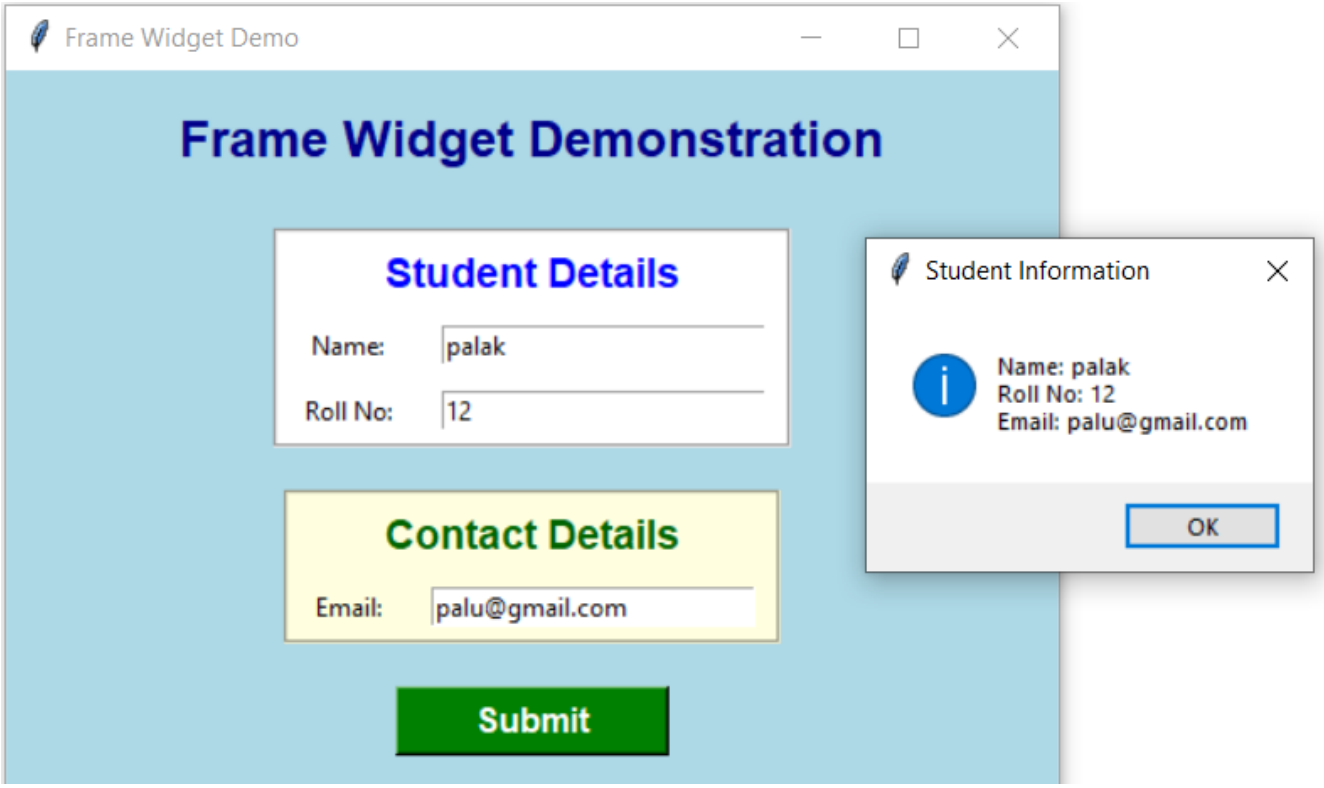
    messagebox.showinfo(
        "Student Information",
        f"Name: {name}\n"
        f"Roll No: {roll}\n"
        f"Email: {email}"
    )

# Submit Button
Button(
    root,
    text="Submit",
    font=("Arial", 12, "bold"),
```

```
    bg="green",
    fg="white",
    width=12,
    command=submit
).pack(pady=10)

# Run Application
root.mainloop()
```

➢ **Output :-**



Q – 11 Write a code to demonstrate Label Widget

- **Input :-**

```
from tkinter import *

# Main Window
root = Tk()
root.title("Label Widget Demo")
root.geometry("450x350")
root.configure(bg="lightblue")

# Heading Label
heading = Label(
   root,
   text="Label Widget Demonstration",
   font=("Arial", 18, "bold"),
   bg="lightblue",
   fg="darkblue"
)
heading.pack(pady=20)

# Simple Label
label1 = Label(
   root,
   text="Welcome to Python Tkinter",
   font=("Arial", 14),
   bg="white",
   fg="black"
)
label1.pack(pady=10)

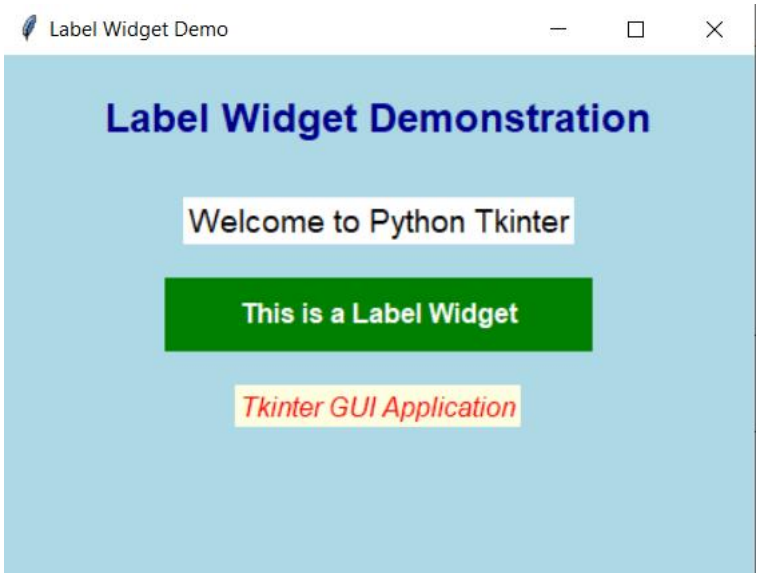
# Colored Label
label2 = Label(
   root,
   text="This is a Label Widget",
   font=("Arial", 12, "bold"),
   bg="green",
```

```
    fg="white",
    width=25,
    height=2
)
label2.pack(pady=10)

# Another Label
label3 = Label(
    root,
    text="Tkinter GUI Application",
    font=("Arial", 12, "italic"),
    bg="lightyellow",
    fg="red"
)
label3.pack(pady=10)

# Run Application
root.mainloop()
```

- **Output :-**



Q – 12 Write a code to demonstrate Listbox Widget

- **Input:-**

```
from tkinter import *
from tkinter import messagebox

# Main Window
root = Tk()
root.title("Listbox Widget Demo")
root.geometry("400x400")
root.configure(bg="lightblue")

# Heading Label
heading = Label(
   root,
   text="Listbox Widget Demonstration",
   font=("Arial", 16, "bold"),
   bg="lightblue",
   fg="darkblue"
)
heading.pack(pady=20)

# Subject Label
label = Label(
   root,
   text="Select Your Subject:",
   font=("Arial", 12),
   bg="lightblue"
)
label.pack(pady=5)

# Create Listbox
subject_list = Listbox(
   root,
   width=25,
   height=6,
   font=("Arial", 12),
```

```
    selectmode=SINGLE,
    exportselection=False
)
subject_list.pack(pady=10)

# Insert Items into Listbox
subjects = [
    "Python",
    "Java",
    "C Programming",
    "Web Development",
    "Database",
    "Computer Network"
]

for subject in subjects:
    subject_list.insert(END, subject)

# Submit Function
def submit():
    selected = subject_list.curselection()

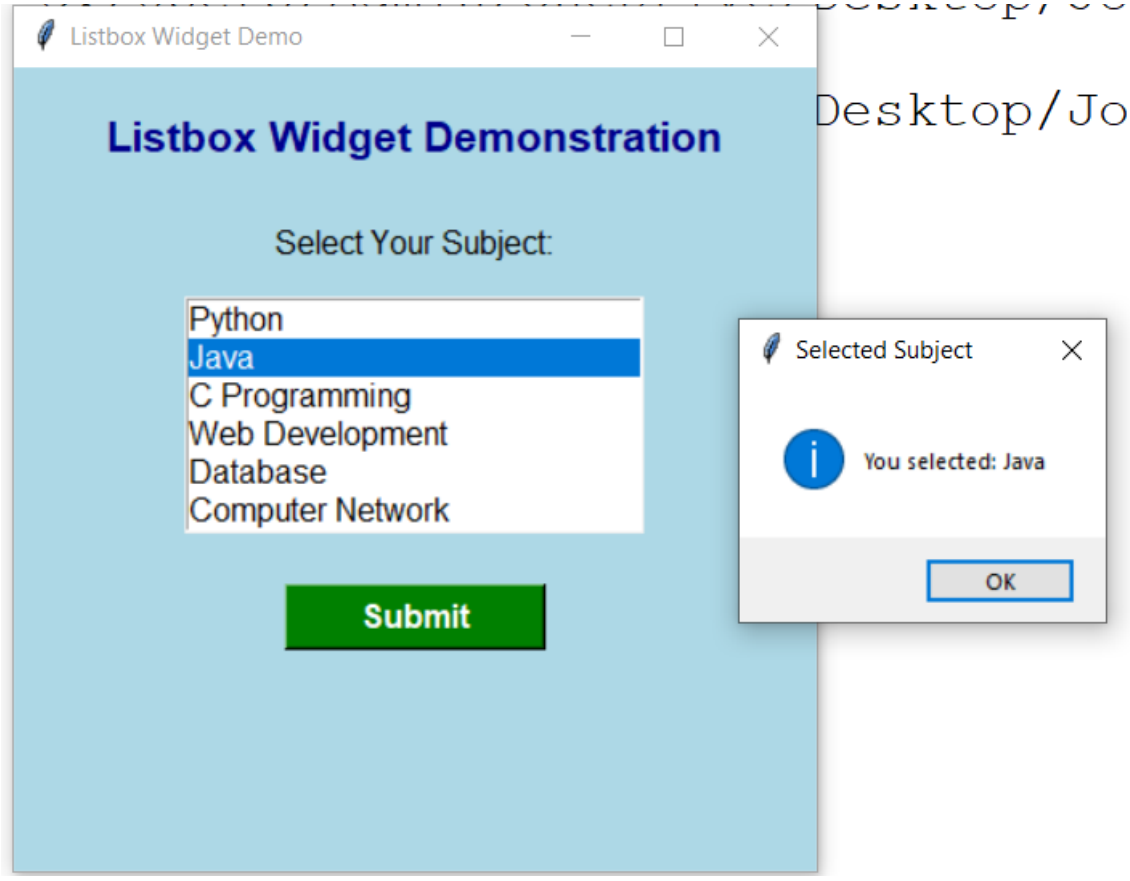
    if selected:
        subject = subject_list.get(selected[0])
        messagebox.showinfo(
            "Selected Subject",
            f"You selected: {subject}"
        )
    else:
        messagebox.showwarning(
            "Warning",
            "Please select a subject."
        )

# Submit Button
btn = Button(
    root,
    text="Submit",
```

```
    font=("Arial", 12, "bold"),
    bg="green",
    fg="white",
    width=12,
    command=submit
)
btn.pack(pady=15)

# Run Application
root.mainloop()
```

➢ **Output:-**



Q – 13 Write a code to demonstrate Radiobutton Widget in Tkinter.

- **Input:-**

```
from tkinter import *
from tkinter import messagebox

# Main Window
root = Tk()
root.title("Radiobutton Widget Demo")
root.geometry("400x350")
root.configure(bg="lightblue")

# Heading Label
heading = Label(
   root,
   text="Radiobutton Widget Demonstration",
   font=("Arial", 15, "bold"),
   bg="lightblue",
   fg="darkblue"
)
heading.pack(pady=20)

# Gender Label
label = Label(
   root,
   text="Select Your Gender:",
   font=("Arial", 12),
   bg="lightblue"
)
label.pack(pady=10)

# Variable for Radiobuttons
gender_var = StringVar(value="Male")

# Male Radiobutton
rb1 = Radiobutton(
   root,
   text="Male",
   variable=gender_var,
```

```
    value="Male",
    font=("Arial", 12),
    bg="lightblue"
)
rb1.pack(anchor="w", padx=130, pady=5)

# Female Radiobutton
rb2 = Radiobutton(
    root,
    text="Female",
    variable=gender_var,
    value="Female",
    font=("Arial", 12),
    bg="lightblue"
)
rb2.pack(anchor="w", padx=130, pady=5)

# Other Radiobutton
rb3 = Radiobutton(
    root,
    text="Other",
    variable=gender_var,
    value="Other",
    font=("Arial", 12),
    bg="lightblue"
)
rb3.pack(anchor="w", padx=130, pady=5)

# Submit Function
def submit():
    gender = gender_var.get()

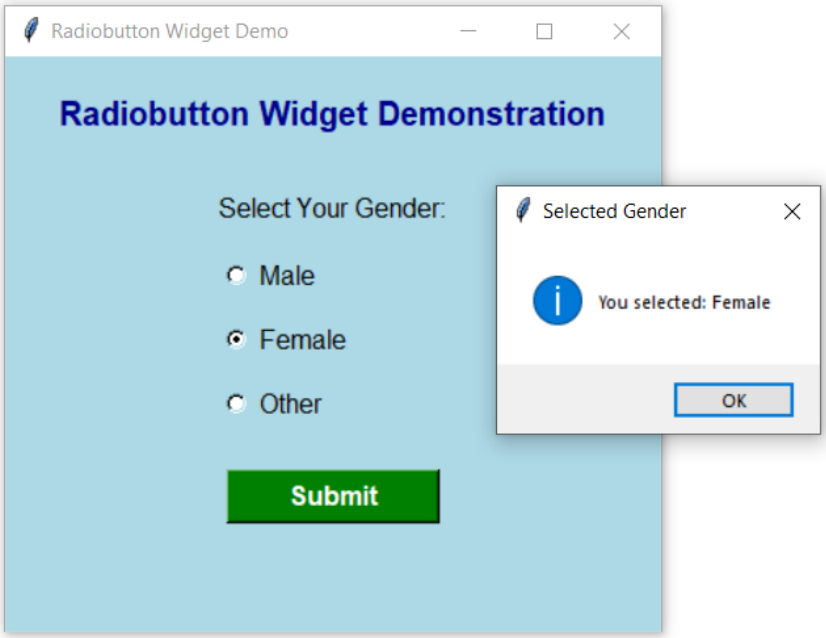
    messagebox.showinfo(
        "Selected Gender",
        f"You selected: {gender}"
    )

# Submit Button
```

```
btn = Button(
    root,
    text="Submit",
    font=("Arial", 12, "bold"),
    bg="green",
    fg="white",
    width=12,
    command=submit
)
btn.pack(pady=20)

# Run Application
root.mainloop()
```

➢ **Output:-**



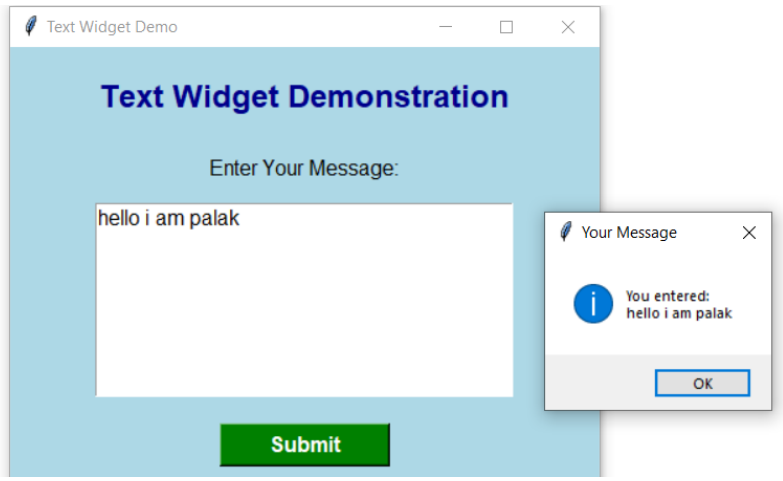
Q – 14 Write a code to demonstrate Text Widget in Tkinter.

- **Input:-**

```
from tkinter import *
from tkinter import messagebox
# Main Window
root = Tk()
root.title("Text Widget Demo")
root.geometry("450x400")
root.configure(bg="lightblue")
# Heading Label
heading = Label(
   root,
   text="Text Widget Demonstration",
   font=("Arial", 18, "bold"),
   bg="lightblue",
   fg="darkblue"
)
heading.pack(pady=20)
label = Label(
   root,
   text="Enter Your Message:",
   font=("Arial", 12),
   bg="lightblue"
)
label.pack(pady=5)
# Create Text Widget
text_box = Text(
   root,
   width=35,
   height=8,
   font=("Arial", 12),
   wrap=WORD
)
text_box.pack(pady=10)
# Submit Function
def submit():
   message = text_box.get("1.0", END).strip()
   if message:
```

```
        messagebox.showinfo(
            "Your Message",
            f"You entered:\n{message}"
        )
    else:
        messagebox.showwarning(
            "Warning",
            "Please enter your message."
        )
# Submit Button
btn = Button(
    root,
    text="Submit",
    font=("Arial", 12, "bold"),
    bg="green",
    fg="white",
    width=12,
    command=submit
)
btn.pack(pady=10)
# Run Application
root.mainloop()
```

➢ **Output:-**



Q – 15 Write a code to demonstrate Scale Widget in Tkinter.

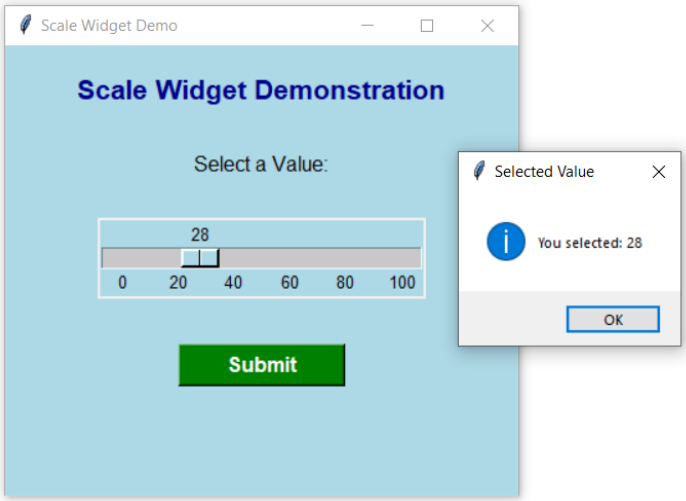
- **Input:-**

```
from tkinter import *
from tkinter import messagebox

# Main Window
root = Tk()
root.title("Scale Widget Demo")
root.geometry("400x350")
root.configure(bg="lightblue")
heading = Label(
   root,
   text="Scale Widget Demonstration",
   font=("Arial", 16, "bold"),
   bg="lightblue",
   fg="darkblue"
)
heading.pack(pady=20)
label = Label(
   root,
   text="Select a Value:",
   font=("Arial", 12),
   bg="lightblue"
)
label.pack(pady=10)
scale = Scale(
   root,
   from_=0,
   to=100,
   orient=HORIZONTAL,
   length=250,
   resolution=1,
   tickinterval=20,
   bg="lightblue",
   font=("Arial", 10)
)
scale.pack(pady=20)
# Submit Function
```

```
def submit():
    value = scale.get()
    messagebox.showinfo(
        "Selected Value",
        f"You selected: {value}"
    )
btn = Button(
    root,
    text="Submit",
    font=("Arial", 12, "bold"),
    bg="green",
    fg="white",
    width=12,
    command=submit
)
btn.pack(pady=15)
# Run Application
root.mainloop()
```

- **Output:-**



Q – 16 Write a code to demonstrate Toplevel Widget in Tkinter.

- **Input:-**

```
from tkinter import *
from tkinter import messagebox

root = Tk()
root.title("Main Window")
root.geometry("400x300")
root.config(bg="lightblue")

def open_window():
   new_window = Toplevel(root)
   new_window.title("New Window")
   new_window.geometry("300x200")
   new_window.config(bg="lightyellow")

   Label(
      new_window,
      text="Welcome to Toplevel Window!",
      font=("Arial", 12),
      bg="lightyellow"
   ).pack(pady=20)

   Button(
      new_window,
      text="Close Window",
      command=new_window.destroy
   ).pack(pady=10)

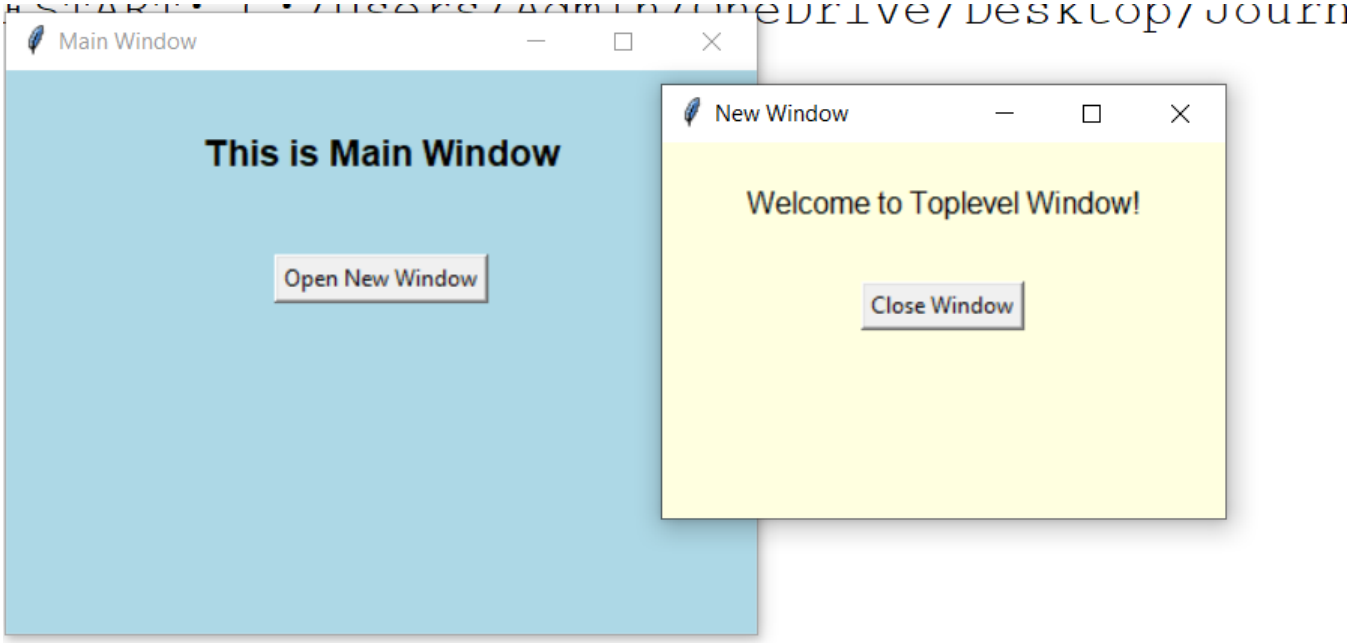
Label(
   root,
   text="This is Main Window",
   font=("Arial", 14, "bold"),
   bg="lightblue"
).pack(pady=30)

Button(
   root,
   text="Open New Window",
```

```
    command=open_window
).pack(pady=10)

root.mainloop()
```

- **Output:-**



Q – 17 Write a code to demonstrate simple calculations in Tkinter.

- **Input:-**

```
from tkinter import *
from tkinter import messagebox
root = Tk()
root.title("Simple Calculations")
root.geometry("400x400")
root.config(bg="lightblue")
def calculate(operation):
  try:
    num1 = float(entry1.get())
    num2 = float(entry2.get())
    if operation == "+":
      result = num1 + num2
    elif operation == "-":
      result = num1 - num2
    elif operation == "*":
      result = num1 * num2
    elif operation == "/":
      if num2 == 0:
        messagebox.showerror("Error", "Cannot divide by zero")
        return
      result = num1 / num2
    result_label.config(text="Result: " + str(result))
  except ValueError:
    messagebox.showerror(
      "Error", "Please enter valid numbers"
    )
Label(
  root,
  text="Simple Calculator",
  font=("Arial", 18, "bold"),
  bg="lightblue"
).pack(pady=15)
Label(root, text="Enter First Number:", bg="lightblue").pack()
entry1 = Entry(root)
entry1.pack(pady=5)
Label(root, text="Enter Second Number:", bg="lightblue").pack()
entry2 = Entry(root)
entry2.pack(pady=5)
```

```
Button(root, text="Addition (+)",
    command=lambda: calculate("+")).pack(pady=3)
Button(root, text="Subtraction (-)",
    command=lambda: calculate("-")).pack(pady=3)
Button(root, text="Multiplication (*)",
    command=lambda: calculate("*")).pack(pady=)
Button(root, text="Division (/)",
    command=lambda: calculate("/")).pack(pady=3)
result_label = Label(
  root,
  text="Result: ",
  font=("Arial", 12, "bold"),
  bg="lightblue"
)
result_label.pack(pady=10)
root.mainloop()
```

- **Output:-**

